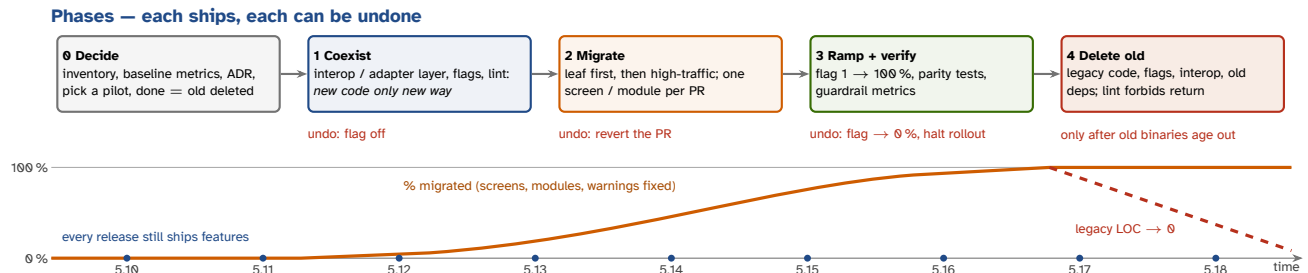


In one line: A large migration (UIKit → SwiftUI, RN Old → New Architecture, Swift 5 → 6) succeeds when **old and new coexist** behind an adapter or interop layer, **new code is written only the new way** from day one, the work moves in **small shippable steps** (each behind a flag or per module, each with a rollback), progress and health are **measured**, and it **ends by deleting the old path**. Feature delivery never stops; a months-long rewrite branch is the failure mode.



How it works

- **Stop the bleeding** first: a lint rule / code-review rule that new code uses only the new way, or the migration target keeps moving.
- **Coexistence layer:** something that lets old and new run in one binary — a hosting controller, RN’s interop layers, per-module Swift language mode, a protocol with two implementations (branch by abstraction). The layer is temporary: it is deleted in phase 4.
- **Order:** a low-risk leaf to learn and write the playbook, then by value and risk; not the hardest screen first, not “alphabetical”. Parity first, redesign later (two hats).
- **Rollback per step:** runtime flag > revert > new build. On mobile old binaries live for months — keep the old path in the binary until the new one is proven, and the server compatible with both.
- **Metrics:** *progress* (% screens/modules, legacy imports, compiler warnings) · *health* (crash-free users, hangs, startup, app size, build time) · *delivery* (features still shipping). A public burndown keeps it funded; a playbook + codemods make the new way the easy way.

Case 1 — UIKit → SwiftUI, screen by screen

- UIKit keeps the **shell and navigation** (coordinators); each migrated screen is a SwiftUI view in a `UIHostingController`. SwiftUI reports intent through closures — it never gets the `UINavigationController`.
- List cells: `UIHostingConfiguration` (iOS 16) inside the existing `UICollectionView`. The other way round: `UIViewRepresentable` for UIKit-only pieces (a text view, a map).
- Shared state: one `@observable` model (iOS 17) or `ObservableObject` read by both sides.
- Order: settings/about → all *new* features → busy lists last; replace navigation with `NavigationStack` last, or never. Snapshot tests prove visual parity.

```

final class ProfileCoordinator {
  let nav: UINavigationController
  func showProfile(_ user: User) {
    let model = ProfileModel(user: user) // @Observable
    let view = ProfileView(model: model,
      onEdit: { [weak self] in self?.showEdit(user) })
    let vc = UIHostingController(rootView: view)
    vc.title = user.name
    nav.pushViewController(vc, animated: true)
  }
}
// a SwiftUI row in an existing UIKit list (iOS 16+)
cell.contentConfiguration = UIHostingConfiguration { RowView(item: item) }
  
```

Remember

“Coexist, stop the bleeding, ship small reversible steps, measure — then delete the old.”

Case 2 — React Native Old → New Architecture

- **Facts:** interop layers since RN 0.74 let many old-architecture libraries run unchanged; New Architecture is the default from Expo SDK 52; **RN 0.82 runs only the New Architecture** (`newArchEnabled=false` is ignored); **0.81 / Expo SDK 54** are the last versions that can still turn it off; SDK 55+ always on.
- **Path** (one variable at a time): upgrade to 0.81 / SDK 54 with the old architecture still on → **audit native deps** (npk expo-doctor against React Native Directory; replace unmaintained ones) → enable the New Architecture on 0.81 → test native modules, custom views, layout, timing → ship with a staged rollout → only then go to 0.82+.
- Your own native code: Codegen specs → TurboModules / Fabric components; the interop layer carries legacy modules meanwhile, but that is a bridge, not a destination.
- **Rollback:** architecture is a *build* setting, not a runtime flag — halt the rollout and ship the previous branch; keep it buildable until the new build is proven.

Case 3 — Swift 5 → Swift 6 language mode

- The language mode is **per module**; a Swift 6 compiler builds Swift 5-mode and Swift 6-mode modules side by side, so migrate target by target.
- Per module: in Swift 5 mode turn on complete concurrency checking and the upcoming features one at a time (*warnings*, builds stay green) → fix → switch that target to Swift 6 (`SWIFT_VERSION = 6` or `.swiftLanguageMode(.v6)`). The guide suggests starting from the outer-most root module. `@preconcurrency import` for unmigrated dependencies.
- Metric: concurrency warnings per module, burned down. Each `@unchecked Sendable` is logged debt.

Traps

- ! A long-lived “v2” branch: merge hell, nothing ships, the old code keeps changing under it.
- ! Migrating and redesigning in the same step — no parity baseline, no clean rollback.
- ! No phase 4: two systems forever (a “lava layer”), double the upkeep.
- ! Treating the RN architecture switch as a runtime flag — it is a build.